

BYLHAH MUGOTITSA

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PHD FELLOW | DATA SCIENCE

Dynamic and results-oriented strategic planner and data scientist with over a decade of experience driving project planning, knowledge management, and continuous analysis in public sector and research settings. I hold a PhD and bring advanced analytics, forecasting, and policy evaluation of health systems—backed by hands-on leadership of large-scale projects such as the INSPIRE Mental Health Data Harmonization Initiative. I use advanced data science (Python, R, SQL), machine learning, and artificial intelligence tools to inform strategic priorities, risk mitigation, and operational efficiency. I have overseen formulating and implementing national and regional data strategies, developed stakeholder-focused performance frameworks, and supported multi-sector resource mobilization efforts.

With strong financial modelling, project management, and programs budgeting capabilities, I align analytics with organizational goals, ensuring data-driven strategic plans, audits, and risk management systems are actionable and sustainable. My work combines rigorous technical knowledge with the ability to synthesize insights and provide high-level advisory support to executive leadership, development partners, and cross-functional stakeholders.

EXPERT SKILLS | HARD SKILLS | TECHNICAL SKILLS | SOFT SKILLS

Strategic Planning & Policy Implementation | Data Standardization & Harmonization | Statistics & Machine Learning | Economic Risk & Analysis | Project & Stakeholder Management | Performance Monitoring & Data Visualization | Natural Language Processing (NLP) | Policy formulation & Implementation.

PROFESSIONAL ACHIEVEMENTS

- Data Science Project Leadership** – Led the Mental Health Data Harmonization Project, developing protocols and risk management plans, and showcasing expertise in managing complex data projects and diverse teams.
- Predictive Data Platform Development** – Directed the creation of a data science platform for mental health data, transforming infrastructure into actionable insights and advancing AI/ML capabilities.
- Project & Stakeholder Management** – Experienced in leading multi-institutional initiatives, mentoring teams, and coordinating technical and advisory support across sectors.
- Data Privacy & Compliance** – Developed compliant analytics platforms to ensure secure, ethical data management, crucial for regulated data handling in projects.
- Stakeholder Collaboration** – Oversaw the appointment of project board members and coordinated with stakeholders and cross-functional teams to align goals, negotiate data agreements, and ensure accountability of project outputs.
- Advanced Data Analytics & Forecasting** – Competent in using R, Python, and SQL for statistical modeling, trend forecasting, and scenario analysis to support strategic choices
- Data Science Recognition & Continuous Learning** – Awarded a PhD scholarship for data science in mental health, reflecting dedication to innovation and advanced AI/ML expertise.

EDUCATION

- 2025 Doctor of Philosophy Health Policy and Management, **Strathmore University, KE | London School of Hygiene & Tropical Medicine, UK**
- 2024 Post Graduate Certificate Data Science and Business Analytics, **University of Texas, United States**
- 2018 Master of Science Applied Statistics, **Jomo Kenyatta University of Agriculture and Technology, Kenya**
- 2015 Post Graduate Diploma Applied Statistics, **Jomo Kenyatta University of Agriculture and Technology, Kenya**
- 2014 Bachelor of Science Financial Engineering, **Jomo Kenyatta University of Agriculture and Technology, Kenya**

PROFESSIONAL VALUE ADDING TRAINING

- Sep 2024 Advanced Epidemiological Analysis, **London School of Hygiene and Tropical Medicine**
- Nov 2023 Systematic Review and Meta-Analysis, **London School of Hygiene and Tropical Medicine**
- Feb 2023 Data Harmonization using Machine Learning, **African Population and Health Research Center**
- Feb 2022 Leadership in Organizations, **UNICAF University**
- Jan 2022 Public Policy Formulation and Implementation, **Kenya School of Government**
- Nov 2020 Senior Management Course, **Kenya School of Government**
- Jan 2020 Training of Trainers, **Kenya School of Government**

AWARDS

Jan 2023	PhD Scholarship (Using Data Science in Longitudinal Research), Wellcome, United Kingdom
Aug 2024	The CODATA-RDA School and Advanced Workshops for Research Data Science, ICTP/UNESCO - The Abdus Salam International Centre for Theoretical Physics
May 2024	Life History Conference Attendance & Presentation, Montreal University & LHC Secretariate

A DECADE OF PROFESSIONAL EXPERIENCE

PhD Fellow	Strathmore University LSHTM CODATA
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Jun 2023 – Present

Leads the Mental Health Data Harmonization Project, leveraging **machine learning**, and **data standardization** through the OMOP Common Data Model. Manages **stakeholder collaboration** for data-sharing agreements across the project HDSS sites in *Kenya, Uganda, Tanzania, Ethiopia, and Senegal*, ensuring **FAIRification** of data and **privacy compliance**. Apply **advanced data science and analytics** for longitudinal integration and application of containerization techniques, leads **systematic reviews** and **landscape analyses**. Oversees **project documentation** and real-time dashboard development to support research reproducibility in Africa. Provides **team mentorship** and conducts **training in AI, policy implementation, data anonymization, pre-processing** and **harmonization** to enhance data science capacity among healthcare, data managers, policy makers and data professionals.

Lecturer-Statistics	Kenya School of Government Nairobi
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Sep 2019 – Present

Provide technical guidance for National and County research initiatives, focusing on advancing **statistical analysis** for accurate forecasting and decision-making. Led **collaborative funding efforts** to secure research support and advanced **project development**. Delivered **capacity building** through training programs, risk management, and contingency planning to enhance public sector research methodologies. Developed **course design** on gender-responsive research and policy and conducted **policy reviews** to support data-driven recommendations. Analyzed **COVID-19 data** with statistical techniques, collaborating with epidemiologists on public health strategies. Mentored emerging leaders, fostering expertise in **SaaS tools** and **sentiment analysis**.

Data Analyst/ Tutor	JKUAT University Nairobi & Kisumu
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Jan 2015 – Jun 2018 (3 Years 5 Months)

Developed **financial models** and an **automated budget tracking system**, cutting analysis time by 50% and enhancing decision-making for government budgets. Conducted **policy and performance analysis**, providing insights to refine tax policy and staff development. Ensured **compliance and accountability** by establishing financial regulations and drafting a research framework for the data center. Led **project funding** initiatives, boosting donor support by 10% and implementing **cost-efficient financial models** that reduced project costs by 20%. Published research in bank fragility and survival analysis of HIV mothers to advance **knowledge-sharing**. Improved **budgeting and reporting processes**, reducing report preparation time by 20%.

10+ YEARS LEVERAGING MY EXPERTISE

Capacity Building and Knowledge Transfer in Government Settings: Designed and led training programs for public servants, policymakers, and data professionals. My experience in strengthening institutional capacity within the Kenya School of Government equips me to support public service work on performance contracting, public participation capacity-building, and strategic plan implementation. I foster a learning culture, critical for replicating innovations and managing intellectual assets.

Advanced Data Science for Strategic Planning and M&E: Led transformative projects such as the INSPIRE Mental Health Data Harmonization Project, applying machine learning (ML) to standardize and analyze multi-country datasets. These skills directly support public service mandate for evidence-based planning, performance measurement, and developing innovative strategies to monitor fiscal and socio-economic indicators. I have used predictive analytics to inform strategic direction and assess the efficiency of public sector initiatives—experience critical for leading internal monitoring systems and reviewing budget implementation.

Data Preparation, Cleaning, and ETL for Public Sector Reporting: Designed and optimized cloud-based ETL processes to clean, structure, and harmonize complex data across multiple government sectors. My work with the OMOP Common Data Model (CDM) in health systems can be adapted to public service needs for standardized fiscal data, enabling accurate performance reporting, feasibility analysis, and knowledge management platforms.

Stakeholder Engagement & Knowledge Partnership Development: Successfully built and managed collaborations with Ministries of Health, academic institutions, research networks, and development partners to support cross-functional goals. These experiences are highly relevant for developing policies, service charters, and securing partnerships and funding streams for public service strategic programs.

Data Privacy, Compliance & Quality Assurance: Championed ethical data governance by designing platforms that comply with privacy standards, FAIR principles, and regulatory frameworks. I have ensured compliance across international data-sharing agreements—an essential competency for implementing business continuity plans, internal control systems, and audit processes at the school. My commitment to transparent, accountable, and compliant data use aligns with the constitutional mandate under Article 228.

Predictive Analytics and Budget Policy Insight: Applied predictive analytics and financial modeling in public sector contexts, providing actionable insights for decision making. These skills directly support responsibilities such as reviewing County Fiscal Strategy Papers, analyzing requisitions for fund withdrawals, and monitoring expenditure compliance with national ceilings—contributing to more prudent and efficient utilization of public funds.

Technical Proficiency for Innovation and Reporting Efficiency: Proficient in Python, R, and SQL, with practical experience building interoperable dashboards, modeling tools, and cloud-based repositories that inform fiscal strategy and operational intelligence—skills relevant for developing and maintaining data and documentation systems within the school.

PROJECTS

Building a Data Science Platform for Longitudinal Mental Health Project

Jul 2023 – Present

Directed the **multi-country Mental Health Data Harmonization Project**, developing and implementing a **Theory of Change framework** to support the **integration of open-source data science platforms for reproducibility of mental health data**. As part of the project, I have also **led the team in assessing the impact of climate shocks on health service delivery through event-driven analytics**, measuring the impact against **Sustainable Development Goals (SDGs) Indicators** using **containerized approaches**.

One of the **project's novel contributions is the development of custom vocabularies that are Athena compliant**. We are currently in discussions with **SNOMED to standardize these vocabularies** to facilitate the mapping of clinical and population data. Another key innovation is the **staging database**, which acts as an intermediary host for longitudinal data and can be analyzed using an **R-extension in ATLAS**.

The project's **primary output, a data science platform for hosting longitudinal population-based data**, is in the final stages of pilot testing for public rollout. The **open-source dashboard and central catalog** will be used globally to **inform and describe African mental health conditions**, allowing policymakers and researchers to **visualize the prevalence, geographic distribution, and trends** over time for mental health outcomes.

REFERRALS/REFERENCES

Mr. John Mose | Chief Finance Officer, State Department for Transport
john.mose2018sp@gmail.com |

Dr. Linda Maoyi | Head of Data, South African Population Research Infrastructure Network (SAPRIN)
linda.maoyi@mrc.ac.za |